MAHAVITARAN
Maharashtra State Electricity Distribution Co. Ltd.

PRESENTATION

Maharashtra Distribution Reforms

Hon'ble Shri Lokesh Chandra, IAS

Addl. Chief Secretary, Chief Minister, Maharashtra

Chairman & Managing Director, MSEDCL

President, AIDA



Agenda

01 MSEDCL: At a Glance

02 Maharashtra Energy Transition Blueprint

03 Solar Adoption: Distributed RE

04 Leading in Rooftop & Solar Pump Installations

05 Grid Automation & Smart Metering

06 Digital Footprint

07 Restructuring & Demerger

08 Results Visible in Performance

MSEDCL: At a Glance

One of the largest distribution companies

3.4 Crore
Consumers

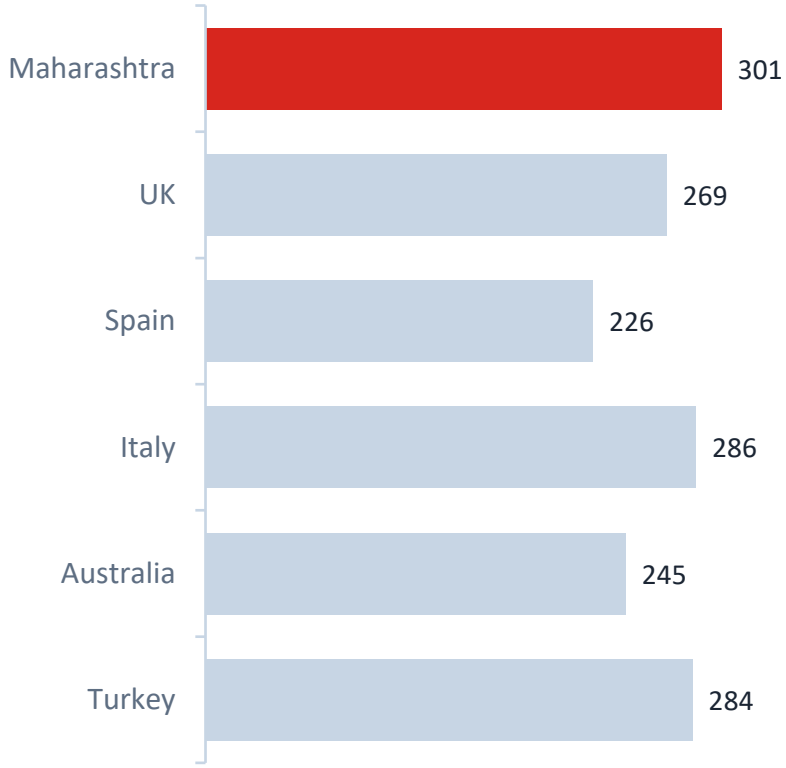
₹1,29,563 Cr
Operation revenue

4,296 nos.
Substations

10 lakh
Distribution transformers

Aggregate and peak demand to grow at 6.3% & 6.5%, reaching ~301K MUs

Demand comparison, thousand MUs



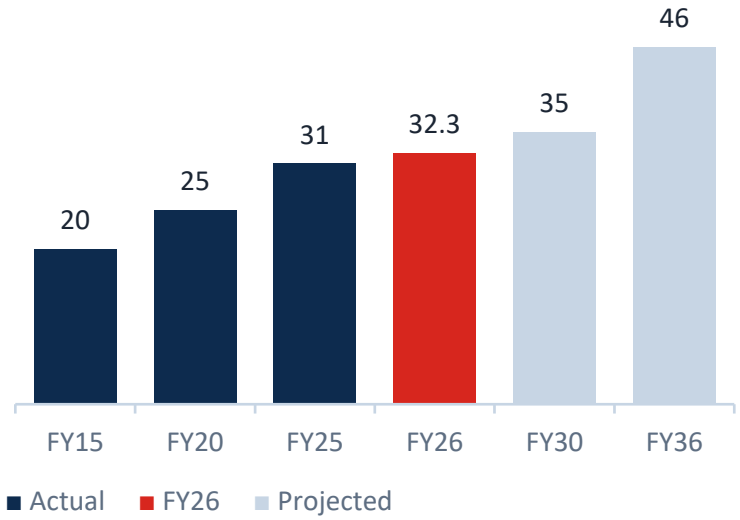
Maharashtra Energy Transition Vision

- India's **leader** in energy transition planning and implementation
- **Fastest acceleration** in capacity, 2023–30
- Facilitating **INR 3 lakh Cr+** investment by 2030; 1/3rd by state utilities
- **Innovation leader** in agri-solarisation, storage & green hydrogen
- **Turn-around** of the distribution company

Energy Transition Blueprint

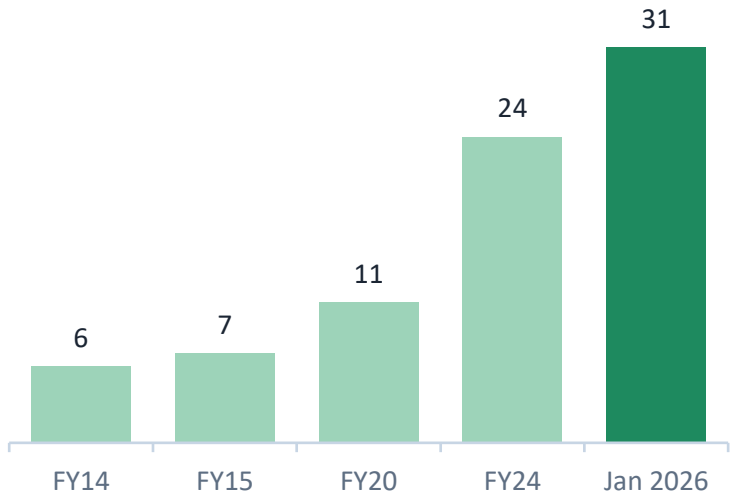
Peak demand: 31 GW (2025) to 46 GW (2036)

Peak demand, GW · 6.5% p.a. growth



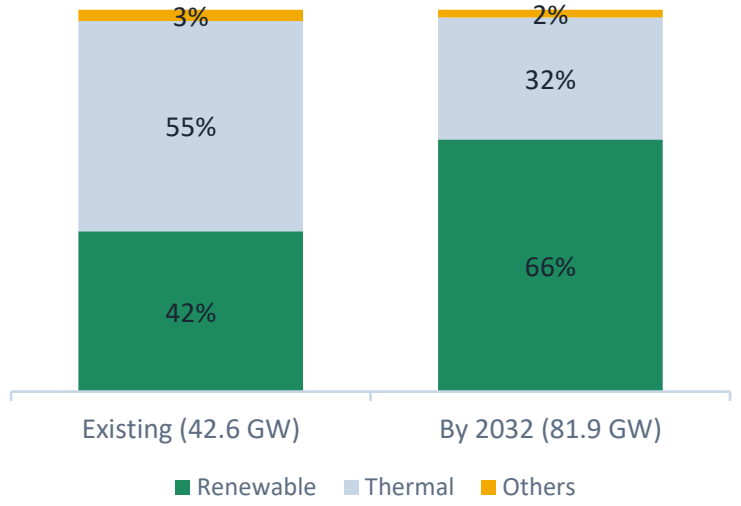
RE capacity up 5 times since FY14

RE capacity, GW (FY14 to Jan 2026)



~40 GW generation to be added by 2032

Capacity mix, % of total GW (as on 31 March)



Key Enablers

7.2 GW
Rooftop solar

16 GW
MSKVY 2.0 distributed solar

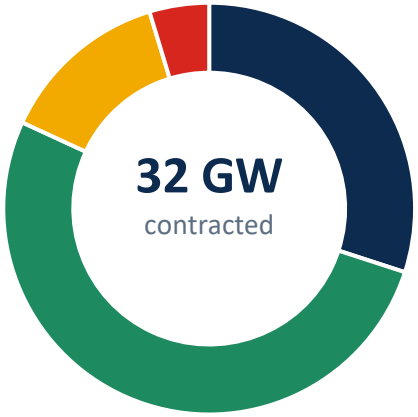
1 Million
Off-grid solar pumps, 1st in India

Contracted capacity **doubled** in 3 years; RE share up from 13% to **52%** by 2030

Renewable: hydro, stand-alone wind, distributed solar (agri-solar), round-the-clock and waste-to-energy capacity. Thermal: coal and gas. Others: nuclear (3% existing, 2% by 2032).

Major Reforms in Distributed Solar

First state to adopt a Resource Adequacy Plan and contract power purchase capacity



- **9.6 GW** — Utility-scale solar
- **16.6 GW** — Distributed agri-solar
- **4.3 GW** — Solar-wind hybrid
- **1.5 GW** — Round-the-clock FDRE

Storage to ease RE intermittency & grid dependence

- **BESS:** 500/1000 MWh commissioned; 2000/4000 MWh awarded
- **PSP:** 28 GWh awarded; MoU for 82 GW with 27 developers
- **RE-RTC contracted:** 2,500 MW

Mukhyamantri Saur Krushi Vahini Yojana

Largest DRE scheme to cater 100% of agri demand

1 Crore	48 Lakh	16,000
Farmers	Agri electricity connections	MW connected load

- Highest agri demand in India: **39,000 MUs**
- Target **16 GW** distributed solar; **~7,000 MW** commissioned
- **38 lakh (80%)** farmers get day-time supply via solar projects & grid

Oct'26 peak demand of **38 GW** met with **32 GW** from grid and **6 GW** from DRE

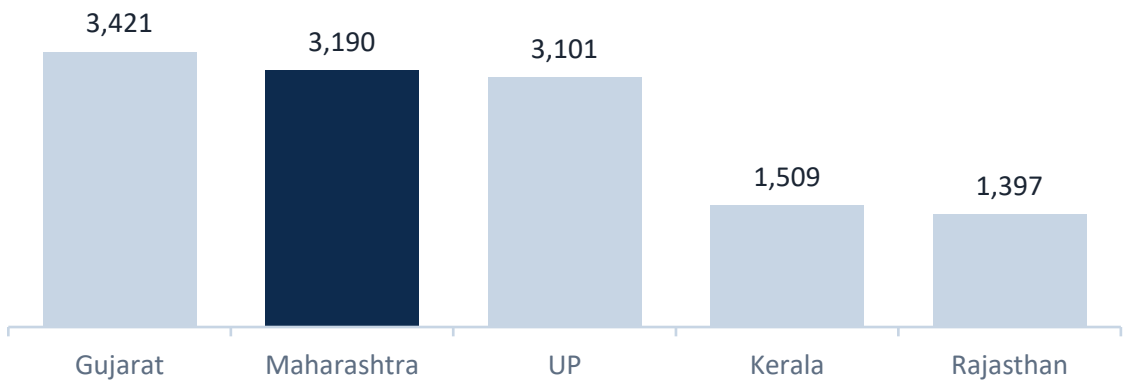
Leader in Rooftop & Solar Pumps

PM Surya Ghar Muft Bijili Yojana

11 lakh+
consumers benefitted; 7,205 MW rooftop solar

₹5,949 Cr
financial support to 7.51 lakh consumers

Installed capacity by state, MW



Achieved **8.54 lakh** installations. FY 2026-27 target: **4.5 lakh** installations (1,800 MW)

Solar Pumps: KUSUM-B & Magel Tyala Saur Krushi Pump Yojana (MTSKPY)

~11 lakh pumps installed in Maharashtra, ~65% of India's installed pumps

5.61 lakh
KUSUM-B

5.2 lakh
MTSKPY

6,285 MW
RE capacity

- **Higher farmer income** from diesel savings; free & reliable power supply
- **Additional target:** 8 lakh pumps by FY30
- **Guinness World Record:** 45,911 pumps installed in a month

Self-Powering Scheme for BPL

SMART: Swayampurn Maharashtra Awasiya Rooftop Scheme

Covers BPL and economically weaker consumers up to **100 units**. Scheme period: **6 Oct 2025 to 31 Mar 2027**.

₹655 Cr
Financial outlay

5 lakh
Beneficiaries covered

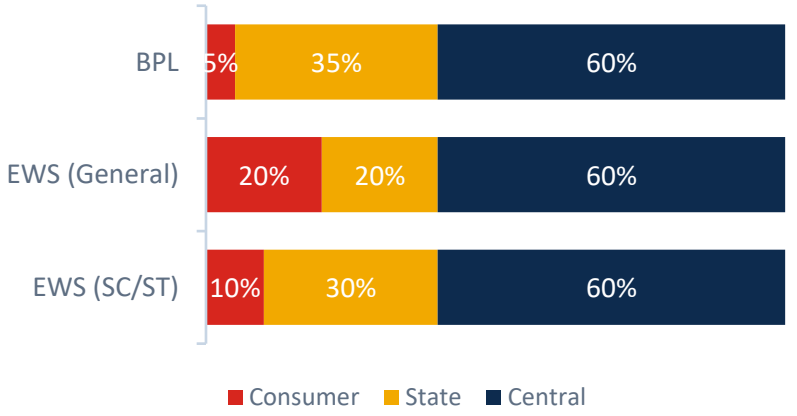
₹45,500/kWp
Rate discovered

3.96 lakh
Applications received; 1,405 installed (1.405 MW)

Category	Consumer share	State subsidy	Central subsidy
BPL households	₹2,500	₹17,500 (35%)	₹30,000 (60%)
Economically weaker (General)	₹10,000	₹10,000 (20%)	₹30,000 (60%)
Economically weaker (SC/ST)	₹5,000	₹15,000 (30%)	₹30,000 (60%)

Each system totals ₹50,000: consumer share + state subsidy + central subsidy

Funding split per ₹50,000 system



AI/ML-Driven Field & Automation

Substation Monitoring System

Low-cost, indigenous automation solution

- Real-time monitoring
- Preventive maintenance
- Equipment diagnosis
- Control Centre dashboard
- Feeder demand & energy accounting
- Load balancing

Integration with DRE projects to balance the Duck curve

AI/ML Demand Forecasting & Billing

Historical Demand Data | **Time Factor** | **Special Events** | **Weather Data**

AI/ML Time-Series Forecasting Engine

Univariate & multivariate models

Scenarios formulated from future demand

Optimal schedule within technical minimum, ramp & MOD limits

Real-time cost-optimized dispatch with outage management & Zero Scheduling

Automated billing from SLDC & WRLDC data, no manual intervention

Smart Metering & Analytics

1. Smart metering for consumers
2. Communicable DT & feeder meters with AMI
3. Meter data analytics using AI/ML

Key highlights (FY 2025-26)

0.0017%
of 91.3 lakh smart meter consumers needed bill revisions

2.13%
of 1.77 crore non-smart consumers registered complaints

Consumption growth (FY 25-26 vs FY 24-25)

+27.2%
Industrial
87.58 → 111.43 MU

+9.6%
Commercial
46.69 → 51.16 MU

+3.0%
Residential
315.2 → 324.7 MU

AMI: Advanced Metering Infrastructure. DT: distribution transformer.

Programme status

₹14,547 Cr
Sanction cost

₹2,182 Cr
Govt. based support

Consumer smart meters
1,38,14,633 installed of 2,41,92,399 sanctioned
57%

DT smart meters
3,67,087 installed of 4,03,862 sanctioned
91%

Digital Footprint: Faceless Services

>3 Cr	2.7 Cr	87%	16,632	3,207
Consumers billed monthly	Registered mobile numbers	Collections via digital channels	Collection counters	Substations monitored

Commercial Operations

- Centralized, integrated billing & collection
- Mobile-first strategy
- Proprietary metering communication protocol

Consumer Services

- Centralized call center
- SMS / e-mail / WhatsApp alerts
- Online payment services
- AI-powered Urja Chatbot (website & app)

Performance Monitoring

- Real-time graphical dashboards
- Operational target setting & monitoring
- Command Center at Prakashgad, Mumbai

Employee Initiatives

- IT-enabled HR processes
- Employee access via website & app
- Paperless approvals for all HR processes

Enabling Systems

- Integrated IT backbone
- AI/ML-based load forecasting
- Substation Monitoring System
- Maintenance portal & GIS

Vendors

- E-tendering portals
- Digital vendor registration, procurement & payments
- Land Portal for MSKVY 2.0

Smart-meter mobile app: **50 lakh+ active users** | MSEDCL is the **first utility to introduce Aadhaar-based OTP authentication**

Restructuring & Demerger

The demerger ring-fences the subsidised agricultural business

MSEDCL

Non-agricultural distribution

- Domestic, commercial & industrial consumers
- Retains distribution network, substations & HT assets
- Retains bulk power procurement & grid operation

MSAPL

Agricultural distribution

- MSEB Solar Agro Power Ltd, subsidiary of MSEB Holding Co. Ltd (incorporated May 2023)
- Already State Nodal Agency for MSKVY 2.0
- ₹2,500 Cr initial capital with state guarantee

1 Bifurcation — ✓ Complete

- Cabinet approval for separate agri company: 7 Apr 2026
- MERC approved licensee: Jun 2026

2 Balance sheet — ✓ Complete

- ₹32,679 Cr of state-guaranteed debt undertaken by GoM

MSEDCL is moving towards becoming **India's first state-owned listed distribution company**

Strategic Gain: Strong Performance

Power purchase savings

~₹82,000 Cr

over the next five years

Low-cost RE procurement rising from **13% to 52%**

Reduced ACoS

₹9.45 → ₹8.56

per kWh, FY25 to FY30 (BAU: ₹14.67 at 9.19% CAGR)

FY26 ACS-ARR gap: **₹-0.54/kWh**

Daytime supply for farmers

38 lakh

farmers (80%) get day-time supply

Daytime electricity to boost farm production & support the state economy. **Process initiated for a separate AG company**

Value creation & rationalisation

26%

reduction in residential (0-100 unit) tariff vs BAU increase of 6.81%

23.9 / 23.7 hrs

supply availability urban / rural; 98.74% billing complaints resolved within SoP

1.73% CAGR

industrial tariff decline vs BAU rise of 8.04%

~₹68,255 Cr

reduction in govt subsidy burden over the next five years

99.86%

collection efficiency, up from 95.82% in 2 years

13.83%

AT&C losses, down from 19.32% in 2 years

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Thank You

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Annexure: Tariff, BAU vs Approved

Tariff trajectory from FY 2024-25 to FY 2029-30: business-as-usual vs approved

Particulars	Existing	Business as Usual (BAU)		Approved Tariff	
	FY 2024-25	CAGR	FY 2029-30	CAGR	FY 2029-30
HT Industry	10.71	8.04%	15.77	-0.09%	**10.67**
LT Industry - General	10.25	6.53%	14.07	0.79%	**10.67**
LT Industry - Powerlooms	9.29	8.77%	14.15	2.80%	**10.67**
HT Commercial	16.55	7.64%	23.91	3.42%	**19.58**
LT Commercial	14.24	7.59%	20.52	4.07%	**17.38**
Residential 1-100	7.58	8.37%	11.33	-4.99%	**5.87**
Residential 101-300	13.35	10.88%	22.37	-2.14%	**11.98**
Residential 301-500	17.95	11.85%	31.42	1.93%	**19.75**
Residential above 500	19.32	12.46%	34.75	0.59%	**19.90**
LT Agriculture	5.70	12.42%	10.22	-8.52%	**3.65**

*BAU CAGRs shown in **red**; approved tariffs that fall over the period shown in **green**.*